Test Plan and Test Cases

SauceDemo Automated Regression Testing (Playwright + JavaScript)

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Part A: Test Plan

1. Objective

To verify, with automated regression tests, that the main user journeys of the SauceDemo web application (login, product listing and sorting, cart, checkout, logout) work correctly, and to be able to re-run the same checks quickly after any change.

2. Application under test

SauceDemo, a public demo e-commerce site: https://www.saucedemo.com

3. Scope

- In scope: login (valid, invalid, locked, empty fields, access protection), product listing and four sort orders, cart add/remove and contents, checkout form validation and order completion, logout.
- Out of scope: performance, security and accessibility testing; other demo users (problem_user, performance_glitch_user, error_user, visual_user); product detail page; Continue Shopping and Cancel buttons; tax and total amount on the overview page; browsers other than Chromium; mobile devices.

4. Test approach

- Type: functional regression testing, fully automated.
- Tool: Playwright Test with JavaScript (Node.js).
- Design: Page Object Model; one class per page in pages/, one spec file per feature in tests/.
- Positive and negative tests: valid flows (TC01, TC14) and invalid input (TC02 to TC05, TC15 and TC16).
- Assertions: Playwright web-first assertions that wait automatically (5 s timeout), no fixed sleeps.
- Isolation: each test runs in a fresh browser context; each spec logs in with a simple beforeEach, so no test depends on another.

5. Test environment

Item	Detail
Operating system	Windows
Browser	Chromium (headless), installed by Playwright
Runtime	Node.js (LTS), npm
Framework	@playwright/test
Editor / VCS	VS Code; Git and GitHub
Network	Internet access to https://www.saucedemo.com

6. Test data

Data	Value
Valid user	standard_user
Locked user	locked_out_user
Password (all users)	secret_sauce
Products used	Sauce Labs Backpack ($29.99), Sauce Labs Bike Light
Customer details	First name Manreet, last name Gill, postal code 12345

All data is stored in utils/testData.js, so it can be changed in one place.

7. Entry and exit criteria

- Entry: Node.js and dependencies installed (npm install), Chromium installed (npx playwright install chromium), SauceDemo reachable.
- Exit: all 17 test cases executed; no failed tests, or every failure analysed and logged as a defect; HTML report generated.

8. Deliverables

- Source code (GitHub repository), test plan and test cases (this document), HTML test report (playwright-report), project report.

9. Risks and assumptions

- SauceDemo is a third-party site; it may change or be unavailable, which can fail tests that are not real defects.
- The sort tests assume SauceDemo shows 6 products; if the site data changes, TC07 to TC09 may need updating.
- Error message wording is checked; a change in wording will fail the matching test.

10. How to execute

- npm install
- npx playwright install chromium
- npm test (all tests) or npm run test:login, test:products, test:cart, test:checkout, test:logout
- npm run report (open the HTML report)

Part B: Test Cases

Automated in: tests/login.spec.js, products.spec.js, cart.spec.js, checkout.spec.js, logout.spec.js. The Actual result and Status columns are intentionally blank. Fill them in only after running npm test yourself, copying what actually happened (Pass or Fail), including any failures.

ID	Feature	Scenario	Steps	Test data	Expected result	Actual result	Status
TC01	Login	Valid login	1. Open https://www.saucedemo.com 2. Enter username and password 3. Click Login	standard_user / secret_sauce	Inventory page opens (URL contains inventory.html); heading is "Products"		
TC02	Login	Wrong password	1. Open site 2. Enter valid username and a wrong password 3. Click Login	standard_user / wrong_password	Error: "Username and password do not match any user in this service"		

ID	Feature	Scenario	Steps	Test data	Expected result	Actual result	Status
TC03	Login	Locked-out user	1. Open site 2. Enter locked user credentials 3. Click Login	locked_out_user / secret_sauce	Error: "this user has been locked out"		
TC04	Login	Empty username	1. Open site 2. Leave username empty, enter password 3. Click Login	username: (empty) password: secret_sauce	Error: "Username is required"		
TC05	Login	Empty password	1. Open site 2. Enter username, leave password empty 3. Click Login	standard_user / (empty)	Error: "Password is required"		
TC06	Login	Direct access without login	1. Without logging in, open /inventory.html	None	Error: "You can only access '/inventory.html' when you are logged in"		
TC07	Products	Product listing	1. Log in 2. Count the products shown	standard_user / secret_sauce	Exactly 6 products are displayed		
TC08	Products	Sort name A to Z	1. Log in 2. Select "Name (A to Z)" 3. Look at the first product	standard_user / secret_sauce	First product is "Sauce Labs Backpack"		
TC09	Products	Sort price low to high	1. Log in 2. Select "Price (low to high)" 3. Look at the first price	standard_user / secret_sauce	First price is $7.99		
TC10	Cart	Add one item	1. Log in 2. Click Add to cart on the backpack	standard_user / secret_sauce Sauce Labs Backpack	Cart badge shows 1		
TC11	Cart	Add two items	1. Log in 2. Add the backpack 3. Add the bike light	standard_user / secret_sauce Backpack, Bike Light	Cart badge shows 2		

ID	Feature	Scenario	Steps	Test data	Expected result	Actual result	Status
TC12	Cart	Remove item	1. Log in 2. Add the backpack 3. Click Remove	standard_user / secret_sauce Sauce Labs Backpack	Cart badge is no longer shown		
TC13	Cart	View cart contents	1. Log in 2. Add the backpack 3. Click the cart icon	standard_user / secret_sauce Sauce Labs Backpack	Cart page opens (cart.html) and lists only "Sauce Labs Backpack"		
TC14	Checkout	Complete purchase	1. Log in, add backpack, open cart 2. Click Checkout 3. Enter details, click Continue 4. Check overview, click Finish	standard_user / secret_sauce Manreet / Gill / 12345	Overview lists the backpack with item total $29.99; confirmation "Thank you for your order!"		
TC15	Checkout	First name missing	1. Go to checkout info page 2. Leave first name empty, fill others 3. Click Continue	first: (empty) last: Gill, zip: 12345	Error: "First Name is required"		
TC16	Checkout	Postal code missing	1. Go to checkout info page 2. Leave postal code empty, fill others 3. Click Continue	first: Manreet, last: Gill zip: (empty)	Error: "Postal Code is required"		
TC17	Logout	Logout	1. Log in 2. Open the menu 3. Click Logout	standard_user / secret_sauce	Login page is shown; Login button is visible		

Part C: Defects and Observations

Defects

To be completed after the final run of the 17 tests. Write "No defects found" only if every test passed; otherwise list each failure with its test ID and error.

Observations

- The tests depend on the current SauceDemo demo data (6 products, first product after each sort, backpack price $29.99). If the site data changes, TC07 to TC09 and TC14 need updating.
- Add any other observations from your own run here.

Not covered (future work)

- Product detail page, Continue Shopping and Cancel buttons, tax and total on the overview page, sort Z to A and price high to low, removing an item on the cart page, last name validation, and cart reset after an order.
- Other demo users, Firefox and WebKit, and running the tests automatically in CI.